



University of St.Gallen

Using the HSG Beamer template

Slide types, figures, tables, CLI agents, Overleaf

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From insight to impact



Agenda

1. Setup: three commands to a finished deck
2. Slide types: every layout the master defines
3. Figures and tables: real research content
4. Workflows: CLI agents and Overleaf
5. Tips: golden rules and common errors

1. Setup

Three commands to a finished deck

1. Bootstrap a new deck folder:

```
bin/new-deck.sh ~/path/to/my-talk
```

2. Edit `deck.tex`: replace every `TODO`.
3. Compile (3 passes for remember-picture overlays to settle):

```
pdflatex deck.tex  
pdflatex deck.tex  
pdflatex deck.tex
```

Or one shot with `latexmk deck.tex` (uses the bundled `latexmkrc`).

Folder layout produced by new-deck.sh

```
my-talk/  
  deck.tex                <- you edit this  
  beamerthemeHSG.sty      <- theme (do not edit)  
  latexmkrc               <- compile config  
  assets/                 <- every image asset  
    hsg-logo-en.pdf, hsg-badge.png, ...  
  fonts/                  <- vendored Gillius ADF (xelatex/lualatex)  
    GilliusADF-Regular.otf, ...
```

The theme adds `\graphicspath{{./assets/}{./}}` so `\includegraphics{hsg-logo-en.pdf}` resolves whether assets are in `./assets/` or flat next to the deck.

Preamble: set once

```
\documentclass[aspectratio=169]{beamer}
\usetheme{HSG}

\title{My talk}
\subtitle{Optional second line}
\author{Prof. Dr. Author Name}
\institute{Your Institute\\University of St.\,Gallen}
\date{}
\coursefooter{Course or project}
\coverimage{cover-photo.jpg}      % optional hero image

\hsgclosingcontact
  {Prof. Dr. Author Name}
  {author@unisg.ch}

\hsgsetsections{Intro, Methods, Results, Outlook}
```

2. Slide types

The cover slide (*Titel Slide 3*)

```
\begin{frame}[plain]\titlepage\end{frame}
```

Just one line in the body. The cover photo comes from `\coverimage{...}` in the preamble. The HSG green badge is layered behind the photo (the badge corners peek through where the photo's clip cuts off).

Tip: the cover photo's right edge is clipped with a slanted polygon. Pick a photo whose subject sits in the *left* third of the frame so the clip does not crop it.

Agenda (*Agenda layout*)

```
\hsgagenda{  
  \item Introduction  
  \item Methodology  
  \item Results  
  \item Discussion  
  \item Conclusion  
}
```

Use 4–6 items, 1–2 words each. The same items should appear in \hsgsetsections so the sidebar progress matches.

Section advance (*silent, no slide rendered*)

```
\hsgsection{Methods}           % advances the sidebar  
% next slide opens "Methods" with a green sidebar marker
```

`\hsgsection` does NOT render a divider slide. It only updates the left-margin progress strip. Place it immediately before the first content slide of each section.

Standard content slide (*Titel und Content*)

```
\begin{frame}  
\frametitle{Results}  
\framesubtitle{Detection performance}  
\begin{itemize}  
\item \hsgemph{AUC = 0.89} on the held-out split  
\item False-positive rate held at 2%  
\end{itemize}  
\end{frame}
```

`\framesubtitle` renders as a tiny grey breadcrumb above the title (keep it under 6 words).

Two columns (*Titel zwei Content*)

Use the `\hsgtwocol` helper. It aligns the left edge of the columns with the frametitle, unlike Beamer's native `columns` which shifts content into the slide margin.

```
\hsgtwocol{%
  \textbf{Symmetric}
  \begin{itemize}\item Shared key\item Fast\end{itemize}
}{%
  \textbf{Asymmetric}
  \begin{itemize}\item Key pair\item Slow\end{itemize}
}
```

Rule: do NOT colour column headers HSGgreen. Reserve green for inline `\hsgemph`.

Two columns rendered

Symmetric

- Shared key
- Fast

Asymmetric

- Key pair
- Slow

Three columns rendered

Symmetric

- Shared key
- Fast

Asymmetric

- Key pair
- Slow

Hybrid

- Best of both
- Practical default

Three-column layout

```
\begin{columns}[T]  
  \column{0.33\textwidth} ...  
  \column{0.33\textwidth} ...  
  \column{0.33\textwidth} ...  
\end{columns}
```

Asymmetric splits work too: $0.6\text{\textwidth} + 0.4\text{\textwidth}$ for a wide-text + narrow-figure pattern.

Three-column rendered

Confidentiality

- Encryption hides content
- Adversary sees only ciphertext

Integrity

- MAC detects tampering
- Hash chains for logs

Availability

- Replication, redundancy
- Rate-limiting against DoS

Three balanced columns with [T] top-alignment. The CIA triad fits naturally into this layout.

Statement (*Statement 1/2*)

```
\hsgstatement{Security is a process, not a product.}
```

Renders at 36 pt HSG green bold on its own slide. Keeps the standard footer chrome.
Use sparingly.



Security is a
process, not a
product.



Quote slides (*Zitat schwarze / weisse Schrift*)

Black-on-white:

```
\hsgquote{The cloud is just somebody else's computer.}{Anonymous}
```

White-on-green:

```
\hsgquoteinverted{The best way to predict the future is to invent it.}{Alan  
Kay}
```



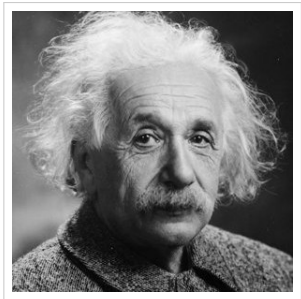
The cloud is just somebody else's computer.

Anonymous



The best way to predict the future is to invent it.

Alan Kay



*Imagination is more important
than knowledge.*

Albert Einstein



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Full-bleed image (*Bild Vollfläche*)

```
\hsgfullimage{path/to/photo.jpg}
```

Image fills the whole 16×9 canvas; small HSG logo overlays bottom-left. Pre-crop the image to 16:9 to avoid letterboxing.

3. Figures & tables

Including a figure

```
\begin{frame}  
\frametitle{Campus reflecting pool}  
\begin{center}  
\includegraphics[width=0.7\linewidth, height=4.6cm,  
keepaspectratio]{hsg-closing-campus.jpg}  
\end{center}  
\end{frame}
```

Always set *both* width and height so the figure fits in the content area without overflowing the footer.

Figure rendered



Caption text sits in the body, not in a figure float. Keep it to one or two lines.

Two figures side by side

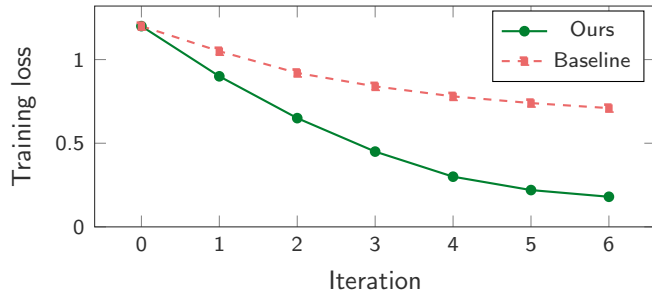
```
\begin{columns}[T]
\column{0.48\textwidth}
\includegraphics[width=\linewidth]{before.jpg}
\scriptsize\itshape Baseline.
\column{0.48\textwidth}
\includegraphics[width=\linewidth]{after.jpg}
\scriptsize\itshape Proposed.
\end{columns}
```

This covers the master's *Titel und zwei Bilder* layout.

Embedding a chart with PGFplots

```
\usepackage{pgfplots}
\pgfplotsset{compat=1.17}
\begin{tikzpicture}
\begin{axis}[width=8cm, height=4.5cm, ymin=0,
xlabel=Iteration, ylabel=Loss]
\addplot[HSGreen, thick] coordinates {
(0,1.2) (1,0.9) (2,0.65) (3,0.45) (4,0.30)};
\end{axis}
\end{tikzpicture}
```

Chart rendered



Tables (booktabs)

```
\usepackage{booktabs}
\begin{tabular}{@{}lrr@{}}
\toprule
Method & AUC & Latency \\
\midrule
Ours & 0.89 & 12 ms \\
Baseline & 0.71 & 8 ms \\
\bottomrule
\end{tabular}
```

Table rendered

Method	AUC	Latency	Params
Ours	0.89	12 ms	4.1 M
Baseline (CNN)	0.71	8 ms	2.3 M
Baseline (LSTM)	0.74	22 ms	3.7 M
Random forest	0.62	3 ms	–

Use booktabs (`\toprule`, `\midrule`, `\bottomrule`) for tables. No vertical lines, no `\hline`.

4. Workflows

CLI agent workflow

An agent (Claude Code, Codex, Cursor, Aider) scaffolds a deck cold:

```
git clone https://github.com/brunobastosrodrigues/beamer-hsg.git
cd beamer-hsg
bin/new-deck.sh ~/talks/2026-conference
cd ~/talks/2026-conference
# fill in TODO markers in deck.tex
pdflatex deck.tex && pdflatex deck.tex && pdflatex deck.tex
```

Three files the agent reads first:

- example/skeleton.tex – canonical starter
- helpers.json – machine-readable API manifest
- AGENTS.md – decision table + golden rules

What helpers.json gives an agent

```
{
  "frame_helpers": [
    {"name": "\\hsgstatement",
      "args": ["text"],
      "layout": "Statement 1",
      "purpose": "One bold green sentence at 36 pt."},
    ...
  ],
  "rules_to_follow": [
    "Three pdflatex passes...",
    "HSG green is reserved for...",
    ...
  ]
}
```

Every public macro: name, arity, master-layout, one-line purpose.

Overleaf workflow

Repo is private, so use the release-ZIP route (no paid Overleaf plan needed):

1. GitHub repo → *Releases* → download *Source code (zip)* for the latest tag.
2. Overleaf → *New Project* → *Upload Project* → drop the ZIP.
3. *Project Settings* → *Compiler*:
 - **XeLaTeX** for the Gill-Sans-Nova look (vendored Gillius ADF picked up automatically).
 - **pdfLaTeX** for the Lato fallback.
4. Open `example/skeleton.tex`, set as *Main document*, replace TODOs, recompile.

Full instructions: `OVERLEAF.md` in the repo root.

5. Tips

Golden rules

- **Never use HSG green for titles or headings.** Reserved for the sidebar, the page badge, and `\hsngemph{...}`.
- **Three passes** of `pdflatex`. Two is not enough; remember-picture overlays drift on pass two.
- **Copy ALL assets** (or use `bin/new-deck.sh`, which does it for you).
- **Keep slides spare.** If you find yourself shrinking to fit, split the slide instead.
- **Cover and closing** are hand-tuned. Do not redefine their templates.

Common errors

Symptom	Where to look
Title chrome misplaced	Compile two more times – remember-picture needs three passes.
Logo or photo missing	An asset under assets/ is missing.
Bullets render green	You added a <code>\setbeamercolor</code> in your preamble. Remove it.
Sidebar progress empty	You did not call <code>\hsgsetsections</code> in the preamble.
Page number clipped	4-digit numbers (1000+) do not fit. Split the deck.

Where to look next

- `AGENTS.md` – exhaustive agent reference + decision table from "what kind of slide do you want?" to "which helper to call".
- `example/skeleton.tex` – minimal compileable starter for any new deck.
- `examples/example.tex` – 9-slide showcase with every helper used at least once.
- `OVERLEAF.md` – step-by-step Overleaf upload.
- `helpers.json` – machine-readable API manifest.

Issues, suggestions, fixes: github.com/brunobastosrodrigues/beamer-hsg.

\hsgtwocol — side-by-side content

Left column

A symmetric two-column body with top alignment and equal gutters. Use it for compare/contrast, figure-plus-caption, or before/after layouts without writing a `minipage` by hand.

Right column

- Top-aligned by default
- 0.48 / 0.48 width split
- Drop *any* content inside

\hsgbibliography — references slide

Place `\hsgbibliography` just before `\hsgclosing`. Requires `biblatex` configured in the preamble:

```
\usepackage[backend=biber, style=numeric]{biblatex}  
\addbibresource{refs.bib}
```

Long lists overflow into extra slides via `allowframebreaks`.

\hsgvideolink — clickable video thumbnail

Drop a 16:9 screenshot next to your .tex and call `\hsgvideolink{url}{thumbnail.jpg}`. The whole figure becomes a hyperlink that opens in the audience's browser.



Panels level themselves

Short

One line.

Longer

- First point
- Second point
- Third point

Medium

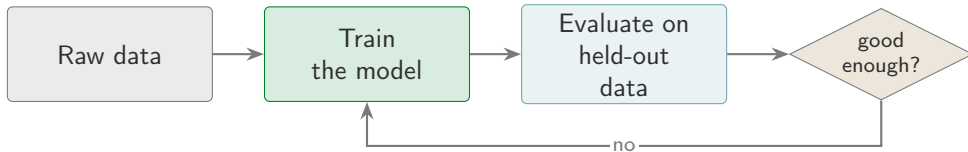
Two lines of text
in this panel.

Definition. `\hsgpanel` without a height, inside `\hsgtwocol` or `\hsgthreecol`, takes the height of the tallest panel in the row.

Takeaway. Rows of panels are level by construction; no height to guess.

`\hsgsource{...}`: the source line, at the bottom, with a gap above.

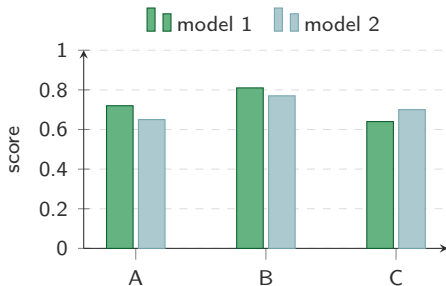
Diagram boxes



Every box has the height of the tallest (three lines), set once on the picture: [hsg box height=1.25cm, hsg box width=2.5cm].

Tables and charts

Metric	What it measures
Accuracy	Share of correct answers
Precision	Share of alarms that are real
Recall	Share of real cases caught
F1	Balance of precision and recall



`hshtable` needs `xcolor=table` in the document class for the stripes; `hsg bars` and `hsg chart` are pgfplots styles.

Animations: `\pause` and `\onslide`

Use Beamer's standard overlay primitives. Each `\pause` step produces a new PDF page, so the page badge increments between clicks — same as any Beamer theme.

- First, set the scene.

Animations: `\pause` and `\onslide`

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- First, set the scene.
- Then, motivate the gap.

Animations: `\pause` and `\onslide`

Use Beamer's standard overlay primitives. Each `\pause` step produces a new PDF page, so the page badge increments between clicks — same as any Beamer theme.

- First, set the scene.
- Then, motivate the gap.
- Finally, the contribution.

For a printable handout, add `handout` to the document class:

```
\documentclass[aspectratio=169,handout]{beamer}
```

Beamer collapses every overlay into a single page automatically.

Backup

Slides for questions and reference, not part of the main narrative.

Backup — Q&A material

Slides after \hsqbackup keep counting toward the page badge but stop inflating the section progress in the top-right corner — it shows *appendix* instead of an extra section number.

Backup — detailed numbers

Use this region for slides you want available during Q&A but not as part of the main narrative.

- Detailed derivations
- Extra results tables
- Limitations or caveats

Questions?



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EFMD
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*"From insight
to impact"* 